What a Protein–Ligand Affinity Model Actually Learns: A Preprocessing Bug, Three Null Ablations, and a Proximity Ceiling on CASF-2016

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Abstract

We report a controlled empirical study of BindFormer, a residue–atom co-attention model that predicts protein–ligand binding affinity (pK) from a complex with a known crystal pose, evaluated for scoring power on CASF-2016. The study is organized around an evaluation protocol built *before* any architectural change: every comparison uses three random seeds and a paired bootstrap over the 285 test complexes, so that an effect the size of a typical architectural tweak (0.01–0.03 Pearson R) can be told apart from seed and test-set noise. The protocol repays its cost immediately. Our single-seed baseline reaches $R = 0.748 \pm 0.011$, but trivial predictors that use no binding information reach $R = 0.50$ (ligand heavy-atom count) and $R = 0.51$ (pocket residue count), so roughly two thirds of the raw correlation is explained by molecular size and pocket identity. A diagnostic baseline then exposed a train/test preprocessing bug—hydrogen atoms were silently retained in the CASF ligands but not the training ligands—whose correction raised Pearson R by about 0.08 everywhere and reversed two conclusions that had otherwise looked significant, including an apparent +0.14 benefit of ESM-2 protein embeddings that corrected to +0.05 at $p = 0.095$. Of four architectural or objective interventions, three produced no significant change on CASF and one significantly hurt, each with an accompanying diagnostic confirming the intervention was mechanistically active rather than inert. A pose-perturbation analysis explains the pattern: rotating the ligand in place costs only 0.02–0.03 Pearson R, while translating it out of the pocket drives the score to zero. BindFormer scores *composition and proximity*, not interaction geometry, and refining how attention weights distances cannot help a model that does not read fine geometry. We offer the protocol, the diagnostics, and the preprocessing lesson as transferable, and release all code.

1 Introduction

Machine-learning scoring functions for protein–ligand binding affinity are usually reported as a single correlation coefficient on a benchmark such as the CASF-2016 scoring power set [1]: Pearson R, RMSE, and the residual standard deviation SD over 285 protein–ligand complexes. Published deep models reach roughly 0.80–0.85 Pearson R. A single number on 285 complexes, however, carries a 95% confidence interval of about ± 0.06, which is wider than the improvement most architectural changes claim. Reporting one number per configuration therefore cannot, on its own, distinguish a

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real advance from noise, and the literature's difficulty in reproducing incremental gains is at least partly a measurement problem.

This paper is a case study in treating the measurement as the first-class object. We build BindFormer, a residue–atom co-attention affinity model in the current idiom (ESM-2 pocket embeddings, bidirectional distance-masked co-attention, an auxiliary contact head), and we build an uncertainty harness—multiple seeds plus a paired bootstrap over complexes—before attempting any improvement. We then use that harness to interrogate the model rather than only to rank it.

The study's findings are mostly negative, and deliberately so:

1. A correctly measured baseline reaches $R = 0.748 \pm 0.011$ (three seeds), with a three-seed prediction ensemble at 0.772.

2. Trivial, no-binding predictors reach $R = 0.50$–0.53 on CASF-2016, so most of the raw correlation requires no model of binding at all. This floor is rarely reported and materially changes how a headline number should be read.

3. A diagnostic baseline uncovered a hydrogen-handling bug that silently inflated the CASF ligands relative to the training ligands. Correcting it was worth +0.08 Pearson R, larger than any architectural change we tried, and it reversed two conclusions.

4. Three architectural or objective interventions (a radial-basis distance bias, attention-pooled readout, and a decoy-pose discrimination objective) produced no significant change on CASF; a sharpened attention pool significantly *reduced* it. In each case a mechanism diagnostic confirms the intervention was active.

5. Pose perturbation shows the model is almost insensitive to ligand orientation: rotating the ligand within its pocket costs 0.02–0.03 Pearson R. The model reads proximity and composition, not interaction geometry.

We do not claim a new state of the art. We claim that a disciplined protocol, paired with cheap mechanistic diagnostics, turns a sequence of ambiguous results into a coherent and falsifiable account of what the model has and has not learned—and catches at least one spurious "significant" result and one data bug that a single-number workflow would have shipped.

2 Model and data

2.1 Architecture

BindFormer scores a given crystal pose. Pocket residues within 10 Å of the ligand are encoded with ESM-2 650M per-residue embeddings [3]; ligand heavy atoms receive 74-dimensional atom features together with a learned per-head bias derived from bond-graph shortest-path distances, in the manner of a graph transformer [4]. Bidirectional residue–atom co-attention is restricted to residue–atom pairs within 10 Å. The model has two heads: affinity regression and an auxiliary residue–atom contact head trained against 4 Å labels. The encoders use a custom biased transformer built on scaled dot-product attention rather than `nn.TransformerEncoder`, because the PyTorch fast path turned the per-head bias mask into `NaN` for batch sizes greater than one.

2.2 Data and splits

We use PDBbind v2020 [2] (refined and general sets, 19,443 complexes) for training and validation, and the CASF-2016 core set (285 complexes) as a held-out test set that never enters training or validation. Validation is a seeded random 10% of the non-core data. Featurization crops each protein to its 10 Å pocket, computes contact labels, and caches per-complex tensors; protein features are stored in half precision. Five of the 19,443 ligands fail all parsing fallbacks (corrupt SDF records in the general set, none in CASF) and are dropped, leaving splits of 17,238 train / 1,915 validation / 285 CASF, verified to be disjoint complex-by-complex.

All training was run on a single Apple M5 Max using the Metal Performance Shaders backend. Preprocessing with ESM-2 took about 11 minutes; each training seed took 70–150 minutes depending on machine contention. The full suite comprises 50 unit tests.

3 Evaluation protocol

3.1 The measurement problem

On $n = 285$ complexes, the 95% confidence interval on a single Pearson R near 0.75 is approximately ± 0.06 (bootstrap). The seed-to-seed standard deviation of our baseline is 0.011–0.022. The architectural changes we test are each expected, from the literature, to move Pearson R by 0.01–0.03. All three of these quantities are the same order of magnitude, so a comparison of two configurations by one number each is uninformative by construction.

3.2 The harness

We therefore adopt a fixed protocol for every comparison (`scripts/compare.py`):

- **Seeds.** Each configuration is trained at three seeds into separate directories; we report the per-seed mean and standard deviation.
- **Paired bootstrap.** Two configurations are compared by resampling the 285 *complexes* (10,000 resamples) and recomputing the metric difference on the seed-averaged predictions, both models scored on the same complexes. Pairing removes test-set sampling noise; seeds remove initialization noise.
- **Residual correlation.** Pairing tightens the interval only in proportion to how correlated the two models' residuals are. The tool reports that correlation rather than assuming it: it is ≈ 0.95 for near-identical variants (where pairing roughly halves the interval) and ≈ 0.73 for ESM-2 versus one-hot (where pairing buys little). We report both the paired interval and an unpaired Fisher-z reference.

A supporting tool, `scripts/eval_checkpoints.py`, re-scores a saved checkpoint on a new feature cache without retraining; this is what made the preprocessing-bug correction (Section 4.2) a matter of re-featurizing and re-scoring rather than repeating every run.

4 Results

4.1 Baseline and the no-binding floor

Three ESM-2 seeds score CASF-2016 at 0.751, 0.758, 0.736 Pearson R (0.748 ± 0.011); the three-seed prediction ensemble reaches 0.772 with a bootstrap interval $[0.705, 0.828]$ and RMSE 1.44. Against

Predictor	CASF Pearson R	RMSE
Ligand heavy-atom count (no fit)	0.500	2.12
Pocket residue count (no fit)	0.508	1.98
Ridge on per-element counts	0.526	1.91
Ridge on mean/sum atom features	0.526	1.91
BindFormer (per seed)	**0.748**	**1.47**

Table 1: No-binding baselines on CASF-2016. Simply counting ligand atoms reaches $R = 0.50$; counting pocket residues reaches 0.51. BindFormer adds about 0.22 over this floor.

the 0.80–0.85 range of published deep models this is a reasonable but unremarkable baseline.

The headline is better read against a floor. Table 1 reports predictors that use no binding information, fitted on the training split and scored on CASF.

Molecular size alone reaches $R = 0.50$, and pocket size 0.51. Roughly two thirds of the raw correlation on CASF-2016 requires no binding model. We report this not to diminish the benchmark but because it reframes every subsequent number: a change that moves Pearson R by 0.02 is moving a quantity whose first 0.50 is molecular bookkeeping.

4.2 A train/test preprocessing bug

The trivial baselines were introduced as a diagnostic, and they immediately exposed a bug. Ridge regression on per-element atom counts scored 0.04—below heavy-atom count's 0.46—which is impossible, since element counts are a superset of the total count. The cause was a single element column with mean 0.11 in training and 13.96 in CASF: hydrogen.

The CASF-2016 core-set SDF files carry explicit hydrogens; the PDBbind general and refined SDFs do not. RDKit's `removeHs=True` preserves hydrogens it deems chemically meaningful rather than stripping them unconditionally, so 36% of cached CASF ligand atoms were hydrogen while the training ligands were heavy-atom only. The test ligands were about 1.6× the length of the training ligands (median 36 versus 23 atoms), and because the pocket crop and the 4 Å contact labels are derived from ligand coordinates, those were computed against hydrogen positions as well. The model was trained on one atom distribution and evaluated on another.

The fix is one call to `RemoveAllHs` on every ligand parse path—the behavior the featurizer's documentation already claimed. Because the training data was almost unaffected ($\approx 11\%$ of complexes lost 1–3 stray hydrogens, verified to preserve finiteness and atom counts), the correction required only re-featurizing the test set and re-scoring the existing checkpoints. It raised CASF Pearson R by about 0.08 for every configuration and reversed two conclusions (Sections 4.3 and 4.4). It was the single largest improvement in the project, larger than any architectural change attempted, which is the study's central cautionary result.

4.3 Protein features: ESM-2 versus one-hot

Replacing ESM-2 pocket embeddings with a one-hot residue-identity encoding, on identical splits, gives the corrected comparison in Table 2.

Measured through the hydrogen bug, this comparison read as $\Delta R = -0.154$ at $p = 0.0002$: an apparently decisive +0.14 Pearson R benefit from ESM-2. The corrupted, oversized CASF ligands harmed the weaker one-hot model (which fell to 0.551) far more than the ESM-2 model (which fell to 0.670), inflating the gap roughly threefold. Corrected, the gap is -0.050 at $p = 0.095$: ESM-2

Protein features	CASF Pearson R	val Pearson R
ESM-2 650M (3 seeds)	0.748	0.741–0.751
one-hot (1 seed)	0.722	0.727

Table 2: Corrected ESM-2 versus one-hot. Paired bootstrap on seed-mean predictions: $\Delta R = -0.050$ $[-0.108, +0.009]$, $p = 0.095$—not significant.

may be worth about 0.05 Pearson R here, but on three seeds it is *not established*. We withdraw the original claim. A related claim—that validation compressed the ESM-2 gap "tenfold"—is also withdrawn; corrected, the gap is 0.019 on validation and 0.026 on CASF, so validation was a fair proxy for this comparison.

One validation finding survives the correction: validation does not rank *seeds*. Across the three baseline seeds, the best validation Pearson R belongs to the worst CASF seed and vice versa; validation Pearson R moves 0.010 across seeds while CASF moves 0.022. Validation remains usable for early stopping but not for choosing among configurations or checkpoints.

4.4 Architectural ablations

We tested two architectural changes that target the model's use of geometry, each on three seeds against the three baseline seeds, each paired on the 285 complexes. For each we computed a diagnostic establishing the mechanism was active, so that a null result means "did not help" rather than "did nothing."

Radial-basis distance bias. Pairwise distances (residue–residue in protein self-attention; residue–atom in co-attention) are expanded on 16 Gaussian radial basis functions and mixed into one additive logit per attention head. The output projection is zero-initialized, so an untrained model is numerically identical to the baseline (asserted to 10^{-6}), and any effect is learned. Result: $\Delta R = -0.003$ $[-0.024, +0.018]$, $p = 0.77$—no effect. The mechanism was active: sweeping distance through the trained modules yields a co-attention bias spanning 1.4–2.6 attention logits, enough to reshape the softmax substantially.

Attention-pooled readout. The affinity head pools residues and atoms with a masked mean; we replaced this with a learned attention pool, again zero-initialized so that uniform weights reproduce the mean exactly. A first attempt produced $\Delta R = -0.004$ at $p = 0.70$, but a diagnostic showed the pooling had never left uniform (attention entropy 0.999 of uniform), making the null uninformative. We then forced the pool to train (an MLP scorer, softmax temperature 0.5, and a 20× learning rate for the scorer), verified on a short run that attention entropy fell to 0.67–0.81, and re-ran. The sharpened pool reaches attention entropy 0.05–0.51 of uniform at the selected checkpoints (the top-weighted token taking 23–58× its uniform share) and scores $\Delta R = -0.032$ $[-0.056, -0.010]$, $p = 0.0076$: a significant *regression*, with RMSE agreeing ($+0.076$, $p = 0.004$).

That mean pooling and near-argmax pooling score within 0.004 of each other—and that forcing selectivity only hurts—indicates the readout is not where the limiting information lives. The optimizer traverses the whole path from uniform to near-argmax on its own and performance is flat along it.

Pose	Pearson R	RMSE	mean $\lvert\Delta\text{pred}\rvert$ (pK)
true	0.748	1.49	—
rotated in place	0.728	1.56	0.27
shifted 2 Å	0.743	1.53	0.28
shifted 5 Å	0.677	1.81	0.66
shifted 10 Å	0.324	3.20	1.63
shifted 25 Å	−0.096	17.77	12.27
scrambled	0.309	3.25	1.60

Table 3: Pose sensitivity on CASF-2016. "Rotated in place" keeps the ligand in its pocket, keeps its centroid, and keeps its internal shape; only which atom faces which residue changes. It costs about 0.02 Pearson R. Translating the ligand out of the pocket destroys the score, but drives the model out of distribution (at 25 Å the predictions are nonsense, not merely wrong), so those rows show the model needs the ligand in the pocket, not that it models interactions.

4.5 What the model uses: pose sensitivity

If the pocket crop and co-attention mask are built from the true pose, a model can score well while ignoring geometry. We test this directly by perturbing the ligand pose at evaluation time, holding all protein and ligand *tokens* fixed so that only the geometry entering the distance mask (and the RBF bias, where present) changes (`scripts/pose_sensitivity.py`). Table 3 reports the sweep, averaged over three seeds.

The informative row is the rotation. Randomizing which atom contacts which residue, while leaving the pocket, the centroid, and the ligand shape untouched, costs only 0.02 Pearson R; a paired bootstrap against the true pose gives -0.018 $[-0.032, -0.004]$ ($p = 0.011$) for a baseline trained on the pre-correction data and -0.032 $[-0.050, -0.016]$ for a baseline trained on corrected data. Either way, atom-level pairing geometry carries almost none of the model's scoring power. The large drops under translation conflate "geometry matters" with "the input left the training distribution," and should not be read as evidence that interactions are modeled.

Together with the trivial floor (Section 4.1), this gives a coherent account: BindFormer scores composition and proximity—the right sort of ligand in the right sort of pocket—and barely uses interaction geometry. This also explains the ablations of Section 4.4: refining how attention weights distances or pools tokens cannot help a model that is not reading fine geometry in the first place.

4.6 A pose-discrimination objective

If affinity regression gives the encoder no reason to use geometry, perhaps an explicit objective will. We added a native-versus-decoy discrimination head whose signal is geometry alone: a decoy shares the native pose's protein, atom features, and bond graph, differing only in the ligand's placement (a random rotation in place, exactly the perturbation the baseline was insensitive to). The decoy's cached attention mask is dropped so the model must rebuild it from the moved coordinates, and a short run confirmed the task is learnable (native-versus-decoy ranking accuracy rising from chance to 76% in three epochs) before committing seeds.

We trained three seeds with the objective and, as a matched control, three baseline seeds on the identical corrected cache—so that the comparison is not confounded by the preprocessing fix. Results in Table 4.

The objective is not significant against either control. Its point estimate is positive (unlike the

Configuration	per-seed CASF Pearson R	seed SD	ensemble
+ decoy objective	0.756	0.022	0.779
baseline, corrected training	0.733	0.017	0.767
baseline, pre-fix training (rescored)	0.748	0.011	0.772

Table 4: Decoy-pose objective. Paired bootstrap on seed-mean predictions: versus the matched control, $\Delta R = +0.012$ $[-0.008, +0.033]$, $p = 0.23$; versus the rescored baseline, $+0.007$, $p = 0.48$. Not significant against either.

Intervention	ΔR (CASF)	p	mechanism verified active
ESM-2 vs. one-hot	+0.050	0.095	—
RBF distance bias	−0.003	0.77	yes (1.4–2.6 logit span)
Attention pool (as built)	−0.004	0.70	no (entropy 0.999)
Attention pool (sharpened)	−0.032	0.008	yes (entropy 0.05–0.51)
Decoy-pose objective	+0.012	0.23	task learned; head unchanged
Hydrogen preprocessing fix	+0.08	—	—

Table 5: All interventions, measured on CASF-2016 by paired bootstrap over 285 complexes, three seeds per arm. The preprocessing fix is not an architectural change and has no paired p; it is the correction underlying every other row's corrected value.

RBF and pooling ablations) and the upper end of its interval (+0.033) would be worth having, so it is the only intervention that might survive more seeds; it is not established on three. More telling is the mechanism check. The pose head learns its task well—ranking accuracy 83.6%, 84.6%, 88.3% at the selected epochs—yet the paired rotation penalty of the affinity head is unchanged: -0.030 $[-0.052, -0.009]$ with the objective versus -0.032 $[-0.050, -0.016]$ without it. The seed with the *best* pose ranking (88.3%) had the *worst* CASF score (0.731). Learning to discriminate poses better did not make the affinity head more pose-aware; the auxiliary task and the regression head are decoupled.

This result also forced a mid-study correction. An earlier reading had compared the pose-trained models' rotation penalty against baselines trained on the pre-correction data (-0.018) and reported the objective as roughly doubling pose sensitivity. The matched baseline on corrected data already sits at -0.032, so the increase is attributable to the hydrogen fix—training on hydrogen-free ligands makes the model read placement more sharply—and not to the decoy objective. We note it as a second instance of the matched control, rather than a convenient one, changing the conclusion.

5 Discussion

5.1 A scorecard of interventions

Table 5 collects the interventions. The one data-preprocessing fix outweighs the four architectural and objective changes combined; three of the four produced no significant effect and one was harmful. The "mechanism active" column is what makes the nulls informative: in every case we can show the intervention did what it claimed, so the null is a statement about the model, not about a dead code path.

5.2 Spurious results caught by the protocol

Three conclusions that a single-number workflow would have reported did not survive the protocol: the +0.14 ESM-2 benefit (an artifact of the preprocessing bug), the claim that validation-based early stopping selects a better CASF checkpoint (within noise, $p = 0.17$), and the claim that the decoy objective doubled pose sensitivity (attributable to the data fix via the matched control). Each was caught by a control or a mechanism diagnostic, not by re-examination of the headline number. This is the practical argument for the protocol: at the noise scale of CASF-2016, cheap controls are not optional rigor but the difference between a result and an artifact.

5.3 Implications for architecture search

Our results suggest that, for a model in this family, searching over attention-level inductive biases is poorly targeted. The model's ceiling is not set by how it weights distances inside attention but by what its training objective rewards: affinity regression on crystal poses rewards reading the pocket and the ligand, and penalizes nothing about getting the geometry wrong. The decoy experiment shows the model *can* be trained to read pose, but as a side task that does not reach the regression head. The productive next step is to couple the geometric signal to the affinity prediction—sharing the pose-discrimination representation with the readout, or weighting the affinity loss by pose confidence—rather than adding attention capacity.

6 Limitations

Several limitations bound these claims. The one-hot ablation uses a single seed, so its comparison, while corrected, has no error bar on the weaker arm. The decoy objective's two controls differ by 0.015 Pearson R per seed (most plausibly noise between independent runs, since the preprocessing fix was verified not to damage the training data), which leaves its effect estimate sensitive to the choice of control; this is itself a reason to regard it as unresolved rather than as evidence of no effect. All evaluation uses the standard random CASF split; we have not run the harder protein-family or LP-PDBBind splits, so we cannot say how much of 0.748 is train/test similarity. The model scores a single crystal pose, and the auxiliary contact head, though trained, is never evaluated against reference interaction fingerprints. Finally, the absolute baseline (0.748) is below published state of the art; the study is about measurement and mechanism, not about topping the benchmark.

7 Conclusion

We built a protein–ligand affinity model and, more importantly, an evaluation protocol to go with it, and used the protocol to ask what the model learns rather than only how high it scores. The protocol paid for itself: it exposed a preprocessing bug worth more than any architectural change we attempted, caught two further spurious conclusions, and turned three null ablations into a single coherent finding—that the model scores composition and proximity, not interaction geometry. We recommend that scoring-power results on CASF-2016 be reported against the no-binding floor ($R \approx 0.50$), with multiple seeds and paired comparisons, and with a mechanism diagnostic for any claimed architectural benefit. The cost is a few extra training runs; the return is the ability to tell a result from an artifact at a noise scale where the two look identical.

Reproducibility. All code, the uncertainty harness, the diagnostic scripts, and the per-seed run records are available at `https://github.com/pankajlp/bindformer-hierarchical-residue-`

`atom-co-attention`. Every quantitative claim in this paper is reproducible from the committed code; `RESULTS.md` in the repository records the raw per-seed numbers and the exact commands. Trained model weights are not redistributed with the code but are available from the author on request; their sha256 sums are committed so any copy can be verified.

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